

Camera Calibration Method under Poor Lighting Condition in Factories

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Abstract: This paper proposes a method to perform accurate camera calibration under poor lighting condition of factories or industrial fields. Preprocessing of camera calibration required for measuring object dimensions has to be able to extract calibration points from patterns of the calibration scale, for example, the calibration from plane pattern scale needs at least seven points of the known dimension marked on the scale. However, industrial fields hardly provide proper lighting condition for camera calibration of the measurement system. The data points for calibration are automatically selected from a probabilistic assumption for size variation of the calibration point when the threshold level changes for image binarization. The system requires user to provide at least four points that are incomplete, these points are used to predict position of exact calibration points and extract accurate calibration parameters in an iterative procedure using nonlinear optimization of the parameters. From real images, we prove the method can be applied to camera calibration of poor quality images obtained under lens distortion and bad illumination.

Keywords: Camera Calibration, Tsai Calibration, Plane Projective Transform, Non-Contact Length Measurement

1. INTRODUCTION

Measuring object and part dimensions in industrial fields is very important because the quality of the products especially depends on precision, accuracy, and reliability of each object part. Popular sensor types for non-contact length measurement includes moire, laser, and camera sensor, but measurement systems by camera is relatively frequently used because it is affordable in prices, easy to install, fast and accurate enough. The camera calibration is essential for measurement of object dimensions by image of camera. It is the process of modeling optical projection through camera lens and relative locations between object and camera in 3D space. Well known methods of camera calibration include Tsai [1] and Zhang [2] algorithms. This paper used Tsai calibration method that provides relatively faster and more accurate results. Tsai method for camera calibration uses precise patterns marked on single or multiple planes and requires the correspondence between 3D coordinates of the pattern marks and their image coordinates from modeling camera internal and external parameters. It requires at least seven calibration points of known position to define all required camera parameters. However, lighting condition in industrial fields is not well-controlled for exact and easy camera calibration, and it is difficult to expect accurate camera calibration results under such bad illumination because the lighting condition disturbs exact extraction of the calibration points.

This paper proposes a calibration method to overcome these limitations. Users are asked to enter coordinates of four points on calibration pattern and then, the system performs perspective transformation from plane pattern of the calibration scale to image plane to estimate location of calibration points on the

image and extract exact calibration points that are distinguished from bad points by poor lighting condition.

Coordinates of the extracted points are used to provide data of the Tsai calibration method and internal and external parameters as result of the calibration are used to extract the points whose distance error between image points and projected points of the calibration scale are smaller than error allowance. By doing so, more calibration points are added to recalculate more accurate camera calibration results on Tsai method. In this iterative process, the calibration method is enough to provide the exact camera parameters at factories that is difficult to control the lighting condition. The method was tested in experiments using real images.

2. CAMERA CALIBRATION

2.1 Plane-to-plane mapping

Fig. 1(a) shows a sample of mechanical part that needs measurement of object dimension by machine vision. A visual inspection system of Fig. 1(b) is equipped with four machine vision cameras whose parameters have to be previously calibrated and LED lamps are attached to control the lighting condition of the inspection sample. This machine is designed to measure objects of long length because it can observe wide view of the object from multiple cameras.

Tsai calibration requires over seven calibration points and users to enter exact coordinates and their correspondence information between calibration points of image and world coordinates. This data preparation becomes a difficult and very inconvenient process if image projection of the pattern scale for camera calibration contains interference under poor lighting condition

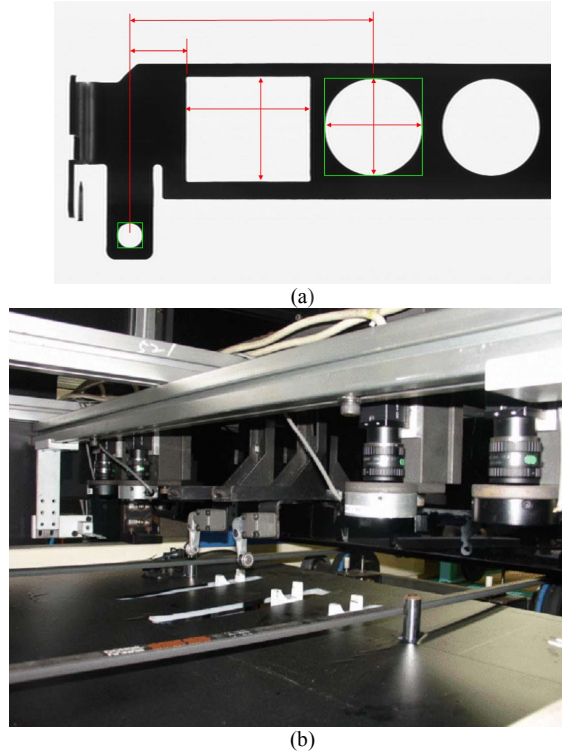


Fig. 1 Dimension measurement of a mechanical part by machine vision system. (a) A sample part; (b) Visual inspection system equipped with FA-cameras

This paper uses the perspective transformation of plane to extract point coordinates for Tsai calibration under bad illumination. A simple perspective projection is plane-to-plane mapping as shown in Fig. 2 that does not include lens distortion [3]. Thus, it is not appropriate for camera calibration. However, as four calibration points can create projection matrix of a plane, it can be applied to initiating Tsai calibration.

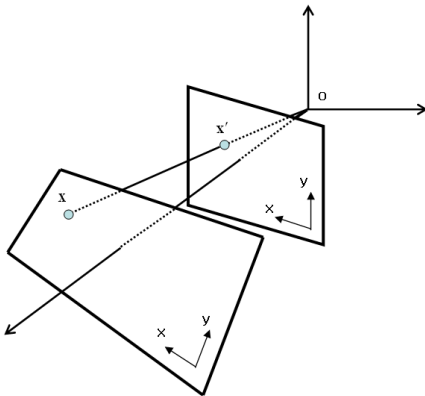


Fig. 2 Plane projective transformation

$$s\mathbf{x}' = s \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{H}\mathbf{x} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (1)$$

Eq. (1) shows the relationship between world coordinates $\mathbf{x}$ of calibration points on pattern scale and calibration coordinates $\mathbf{x}'$ on image plane transformed by homography matrix $\mathbf{H}$. The value s is a scale factor and we can offer $h_{33}=1$ as a constraint to limit magnitude of homography matrix elements. Eq. (1) provides followings:

$$\begin{aligned} x'(h_{31}x + h_{32}y + 1) &= h_{11}x + h_{12}y + h_{13} \\ y'(h_{31}x + h_{32}y + 1) &= h_{21}x + h_{22}y + h_{23} \end{aligned} \quad (2)$$

$$\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -x'x & -x'y \\ 0 & 0 & 0 & x & y & 1 & -y'x & -y'y \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix} \quad (3)$$

Using pseudo-inverse formula from Eq. (3), we can find $h_{11} \sim h_{32}$ for plane projection transformation [4]. Once homography matrix is found, the world coordinates on calibration pattern scale can be projected on the image plane. Then, points with small distance errors between the pattern projection and image points shall be extracted for a nonlinear optimization.

2.2 Selection of data points for calibration

A sample of pattern scale image acquired from a camera is shown in Fig. 3. Tuning the lights such as LED or halogen lamp for uniform intensity on the calibration scale is not easy under dynamic illumination conditions of factories. The image is binarized to intensity values of 0 and 255 as shown in Fig. 3(b) and then image labeling for pixels of 255 is performed. Otsu method can be used to select the threshold value for automatic binarization of the gray-scale image [5]. The method provides an optimal threshold constant to minimize the probabilistic variance for two normal distributions of background and object in image region. Labeled areas are mixed with noisy regions and accurate calibration points as shown in Fig 3(b). The blob

regions of small size are remained after the large size blobs are eliminated. The calibration points of area marked with dotted circle in Fig. 3(c) are on the regions of intensity value similar to the threshold value for binarization. Because boundaries of the blob points may include the background regions and so center position of the regions can shift and distort, we have to remove the blobs of unclear boundary.

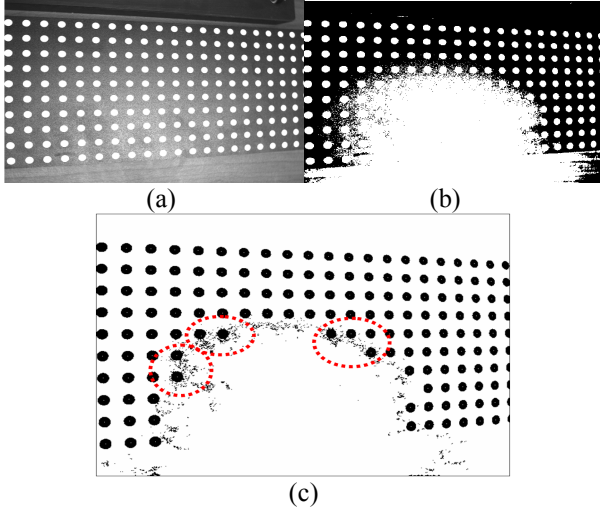


Fig. 3 Extraction of calibration points. (a) Original image; (b) Binarization; (c) Blob areas. The regions marked with dotted circles indicate the regions of sensitive blob size when the threshold level changes

Fig. 4 shows the binarization example for two different areas on a calibration scale. The rectangle area of left side has two clear intensity concentrations of background and point regions in the histogram graph as shown in left below side of Fig. 4. But the area of right side shows a histogram of intensity values mixed for background and point areas. When we change a small value of the threshold, size of blob regions on left side is not change or change at small range, however, the blob size of right side changes very large. If we assume a probabilistic distribution for the changing blob value of uniform lighting area, the random value d for changing size of blob region can be defined as a normal distribution with zero mean and variance σ_d :

$$d \sim N(0, \sigma_d^2) \quad (4)$$

$$\sigma_d = \frac{\sigma_m}{0.68\sqrt{2}} \quad (5)$$

We can calculate the variance according to variation of threshold value for the calibration points only on uniform lighting area. To select automatically the variance without user intervention, variances of all calibration points are sorted by the order of their magnitude and the median value σ_m is selected and σ_d is obtained as shown in Eq. (5). Based on two sigma rule, all blob regions of variance with bigger than

$2\sigma_d$ are removed as points of unstable blob size when the threshold value changes in a small range.

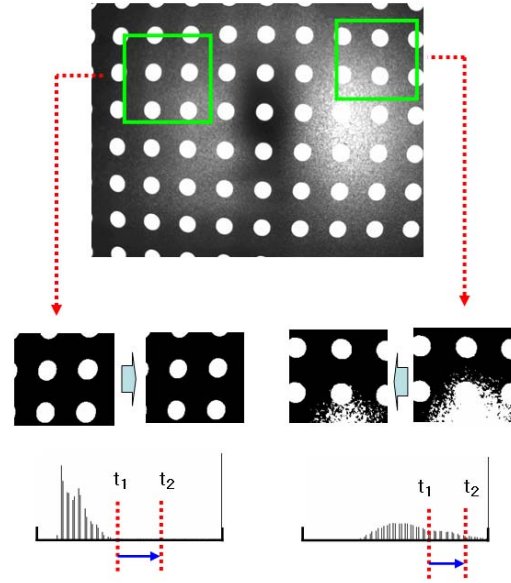


Fig. 4 Comparison for size variation of calibration points on two different lighting areas according to the change of threshold value

Once the selection of calibration points is completed, then we can perform the camera calibration from the remained blob points. Among remained points, the user has to select four calibration points and teaches number of calibration points in two aligned directions. Perspective projection is performed for these four points to derive out the homography matrix. We select only the points with the distance of small value between image projection of 3D points on the scale and central coordinates of labeling points in image. These points are considered calibration points and used as input data into Tsai calibration algorithm. The homography matrix represents only the plane-to-plane mapping relation and does not care lens distortion of the camera system. So there always exists the distance error between the image and the projected points, specifically on image boundary area by the radial distortion of camera lens.

$$|\mathbf{x}' - \mathbf{m}| = \varepsilon < Th_1 \quad (6)$$

If Tsai calibration result that considers the radial distortion is smaller than allowance of the distance error between the image and projected points, the algorithm is ended. If not, internal and external parameters in Tsai calibration shall be used to project the points of the pattern scale into image plane again. Because result of Tsai calibration considers the radial distortion of camera lens, the coordinates of projected points approaches to position of labeled points of the image and so more calibration points are extracted. Extracted coordinates of

more points can be used to reexecute the above calibration process and repeat it until calibration results are smaller than error allowance. Even though bad illumination causes interference, accurate camera calibration results can be derived. Following steps represent the procedure to select calibration points under poor quality lighting condition.

2.3 Camera calibration for bad illumination

Calibration points selected from Part 2.2 shall be used to perform Tsai calibration. Tsai algorithm is the method of calculating internal/external camera parameters and internal parameter is required to define characteristics that relate to camera's optics, geometry, and digital sampling. External parameter defines the geometric transformation between an unknown camera coordinates and world coordinates.

Tsai calibration can accurately express camera distortion and use world coordinate projection to extract more calibration points. This process is repeated until Tsai calibration error is smaller than allowance to optimize camera calibration results.

Table 1 The proposal calibration method for bad illumination

Step 1: Get the image projected from pattern scale for camera calibration
Step 2: Select the positions of four points for plane-to-plane mapping
Step 3: Calculate the plane mapping by using the four points from the plane of calibration pattern to image plane
Step 4: Check the distance error of each projected point.
Step 5: Perform Tsai algorithm with points of small distance error
Step 6: Using the camera and lens distortion parameters of Tsai method, project again the calibration points of target pattern scale to image plane
Step 7: If the calibration error is small enough, then finish the steps, else add the more points to Tsai calibration data and goto Step 5

3. EXPERIMENTS

The experiments used Matrox Meteor II frame grabber board and Sony XC-75 FA camera for image acquisition. The pattern for camera calibration consists of round dots, placed at a distance of 10mm and covering 900mm x 100mm (Fig 5).



Fig. 5 Camera calibration setting

Once homography matrix is found, world coordinates can be projected on the image plane as shown in Fig 6. Then, points with smaller errors shall be extracted as shown in Fig 7. Green color cross notation on Fig 6 represents the center points of labeled areas and red represents projected coordinates by the plane-to-plane mapping. In Fig 7, red color notation represents the calibration coordinates with pixel error smaller than a threshold value that is experimentally decided.

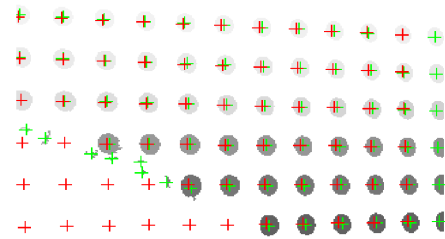


Fig. 6 The result of plane-to-plane mapping from the calibration scale to image plane

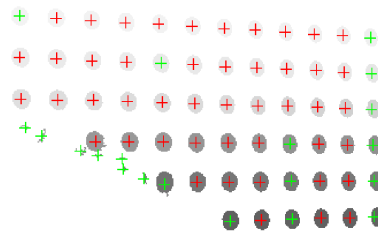


Fig. 7 Extraction of small error points as input data for Tsai calibration

Table 1 represents the camera calibration results P-1 and P-2 with no lighting interference and P-3 and P-4 with lighting interference. It shows the number of calibration points and Tsai calibration error extracted during iteration of the optimization process. Calibration error adversely projected calibration coordinates to average distance to world coordinates [6].

Table 2 Example of camera calibration.

No	ise	□		□		□	
		# of Extraction	Calibration Error	# of Extraction	Calibration Error	# of Extraction	Calibration Error
No	P-1	33	0.128670	130	0.107300	130	0.098331
	P-2	99	0.131076	158	0.108006	160	0.117237
Yes	P-3	70	0.103283	113	0.140141	113	0.134100
	P-4	91	0.170282	188	0.137337	188	0.147177

Fig 8, 9, and 10 show the results. Yellow points are the extracted points, blue points are the world coordinates, and green points are the center of labeled areas.

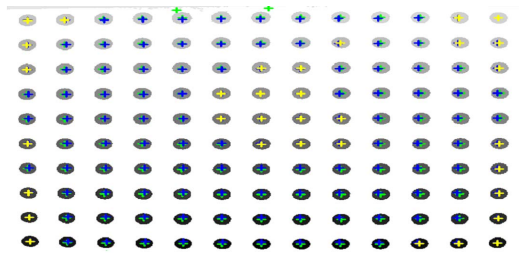


Fig. 8 P-1's 1st Calibration Points

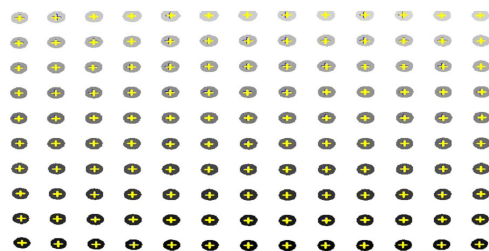


Fig. 9 P-1's Last Calibration Points

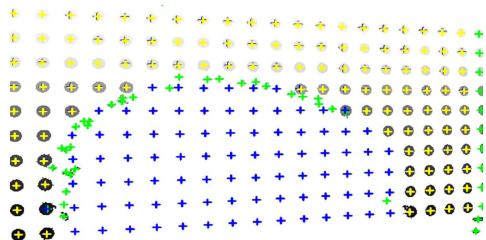


Fig. 11 P-3's Calibration Points

4. CONCLUSION

This paper proposed semi-automation pattern calibration point extraction and optimal camera calibration methods for accurate camera calibration under bad illumination. Noncontact measure can be performed by simple operation of user at industrial sites. The measuring device can be easily handled by experts and non-experts and influence of bad illumination can be minimized to perform accurate camera calibration solely by controlling the brightness of camera lens. This makes measuring convenient in any environment. Later, stereo camera would be used

for 3D measuring, using automatic algorithm instead of semi-automatic algorithm.

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